Introduction to Linear Aljebra:

Matrix

$$A\,adj(A) = Adj(A).A = |A|I$$

Inverse $A^{-1} = \frac{Adj(A)}{|A|}$

Q1. Find inverse of Matrix?

$$A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

B is the inverse of A when

$$AB = BA = I$$

Q. Find the inverse?

$$A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

Ans: $|A| = 1(16-9) - 3(4-3) + 3(3-4)$

$= 7 - 3 - 3$

$= 1$

Adj A =

Cofactor $= (-1)^{i+j} M_{ij} = \begin{bmatrix} 7 & -1 & -1 \\ -3 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$

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$$\text{Adj } A = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{\text{Adj } A}{|A|} = \frac{\text{Adj } A}{1}$$

b) $$A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$

$$|A| = 2(2) = 4$$

Adj A =

Cofactor = $$\begin{bmatrix} 2 & 0 & 0 \\ -6 & 4 & 0 \\ 2 & -4 & 2 \end{bmatrix}$$

$$\text{Adj } A = \begin{bmatrix} 2 & -6 & 2 \\ 0 & 4 & -4 \\ 0 & 0 & 2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1/2 & -3/2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1/2 \end{bmatrix}$$

Linear Equation

An eqⁿ of the form $a_1x_1 + a_2x_2 + \cdots a_nx_n = b$ where a_1, a_2, a_n are the coeff.

eg: $3x + 2y = 4$

Linear System of Eqⁿ

A linear system is a collection of 1 or more linear eqⁿ involving the variables $x_1, x_2 \cdots x_n$.

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$
$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$
$$\vdots$$
$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

→ A linear system can have either a linear sol^n, or no sol^n or infinitely many sol^n.

Representation as Matrix

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

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Row elementary Operations

① Replacement: Replace one row by the sum of itself and a multiple of another row.

② Interchange: Interchange two rows.

③ Scaling: Multiply all the entries in row by a non zero constant.

Row equivalant Matrix

Two matrices are row equivalent in their is sequence of elementary row operation that transform one matrix to another.

Augmented Matrix

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$[A|b] = \begin{bmatrix} a_{11} & \cdots & a_{1n} & b_1 \\ a_{21} & & \vdots & b_2 \\ \vdots & & & \vdots \\ a_{m1} & & a_{mn} & b_m \end{bmatrix}$$

RESULT: If the augumented matrices of two linear systems are row eq then the two system have same solution set.

Q If ordered pair (3,4,-2) is a solution of following system.

$$5x_1 - x_2 + 2x_3 = 7$$
$$15 - 4 - 4$$
$$15 - 8 = 7$$
$$-2x_1 + 6x_2 + 9x_3 = 0$$
$$-6 + 24 - 27$$
$$-9 \neq 0$$
$$-7x_1 + 5x_2 - 3x_3 = -7$$
$$-21 + 20 + 6 = -7$$
$$5 \neq -7$$

Not a Soln

→ Row Echelon Form

A matrix is said to be in row echelon form if it has the following properties:

a) all non zero rows are above any rows of all zeros.

b) Each leading entry of a row is in a column to the right of the leading entry of the row above it.

c) All entries in a column below a leading entry are zeros.

Note: The first non zero element in a row is called leading entry of a row.

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Gaussian Elimination

a. Start with the first column: If it has all entries zero move to the next column to the right.

b. If the column has non zero entries interchange rows if necessary to get a non zero entry on the top.

c. ~~Change the top entry if necessary~~

c. For any non zero ~~bet~~ entry below the top one use an elementary row operation to change it to zero.

d. Now consider the part of the matrix below the top row and to the right of the column under consideration.

e. If their are no such row of column procedure is finished otherwise carryout the same procedure on the new matrix.

Q. a) $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 0 \\ 0 & 2 & 3 \end{bmatrix}$

$R_2 = R_2 - 4R_1$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -12 \\ 0 & 2 & 3 \end{bmatrix}$$

$R_3 = R_3 + \frac{2}{3} R_2$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -12 \\ 0 & 0 & -5 \end{bmatrix}$$

Q.)

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 5 & 7 & 0 \end{bmatrix}$$

$R_2 = R_2 - 2R_1$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 5 & 7 & 0 \end{bmatrix}$$

$R_3 = R_3 - 5R_1$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -3 & -15 \end{bmatrix}$$

after interchanging the rows $R_2 \leftrightarrow R_3$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -15 \\ 0 & 0 & 0 \end{bmatrix}$$

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Exercise 1

Determine which matrice are in row echelon form:

a) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ = Yes

b) $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ = Yes

c) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ = No

d) $\begin{bmatrix} 1 & 1 & 0 & 1 & 1 \\ 0 & 2 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$ = Yes

e) $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ = Yes

f) $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ = Yes

g) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ = No

h) $\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ = Yes

Q.4. $\begin{bmatrix} 0 & 1 & 2 \\ 0 & 4 & 5 \\ 0 & 6 & 8 \end{bmatrix}$

$R_2 = R_2 - 4R_1$ $\begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & -3 \\ 0 & 6 & 8 \end{bmatrix}$

$R_3 = R_3 - 6R_1$ $\begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & -4 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 & 2 \\ 0 & 6 & 8 \\ 0 & 0 & -3 \end{bmatrix}$

Q.5) $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Q.6) $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

$R_2 = R_2 - 4R_1$ $\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{bmatrix}$

$R_3 = R_3 - 7R_1$ $\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix}$

$R_3 = R_3 + 2R_2$ $\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix}$

Q4) $\begin{bmatrix} 0 & 1 & 2 \\ 0 & 4 & 5 \\ 0 & 6 & 8 \end{bmatrix}$

$R_2 = R_2 - 4R_1$

$R_3 = R_3 - 6R_1$

Q6) $\begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}$

$R_1 \leftrightarrow R_2$ $\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}$

$R_2 \leftrightarrow R_4$

$R_4 \leftrightarrow R_3$ $\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Rank: no of non zero rows in row echelon form

H.W

3. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 6 & 7 & 8 & 9 \end{bmatrix}$

$R_2 = R_2 - 4R_1$ $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -6 & -9 \\ 6 & 7 & 8 & 9 \end{bmatrix}$

$R_3 = R_3 - 6R_1$ $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -6 & -9 \\ 0 & -5 & -10 & -15 \end{bmatrix}$

$R_3 = R_3 - \frac{5}{3}R_2$ $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -6 & -9 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

4. $\begin{bmatrix} 1 & 3 & 5 & 7 \\ 3 & 5 & 7 & 9 \\ 5 & 7 & 9 & 1 \end{bmatrix}$

$R_2 = R_2 - 3R_1$ $\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & -4 & -8 & -12 \\ 5 & 7 & 9 & 1 \end{bmatrix}$

$R_3 = R_3 - 5R_1$ $\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & -4 & -8 & -12 \\ 0 & -8 & -16 & -34 \end{bmatrix}$

$R_3 = R_3 - 2R_2$ $\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & -4 & -8 & -12 \\ 0 & 0 & 0 & -10 \end{bmatrix}$

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Q. Solve the following system (Check if its consistent)

$$3x_1 + 5x_2 - 4x_3 = 0$$
$$-3x_1 - 2x_2 + 4x_3 = 0$$
$$6x_1 + x_2 - 8x_3 = 0$$

Sol:-

$$\begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R_2 = R_2 + R_1$

$$\left[\begin{array}{ccc|c} 3 & 5 & -4 & 0 \\ 0 & 3 & 0 & 0 \\ 6 & 1 & -8 & 0 \end{array}\right]$$

$R_3 = R_3 - 2R_1$

$$\left[\begin{array}{ccc|c} 3 & 5 & -4 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & -9 & 0 & 0 \end{array}\right]$$

$R_3 = 3R_3 + 3R_2$

$$\left[\begin{array}{ccc|c} 3 & 5 & -4 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$Ax = b$

Rank $(A|b)$ = Rank A — Rank $(A|b) \neq$ Rank A — non sol?

number of cols = n — $\neq n$

unique — inf

1.5

1) $2x_1 - 5x_2 + 8x_3 = 0$
$-2x_1 - 7x_2 + x_3 = 0$
$4x_1 + 2x_2 + 7x_3 = 0$

$$\begin{bmatrix} 2 & -5 & 8 \\ -2 & -7 & 1 \\ 4 & 2 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R_2 = R_2 + R_1 = \begin{bmatrix} 2 & -5 & 8 \\ 0 & -12 & 9 \\ 4 & 2 & 7 \end{bmatrix}$

$R_3 = R_3 - 2R_1 \begin{bmatrix} 2 & -5 & 8 \\ 0 & -12 & 9 \\ 0 & 12 & -9 \end{bmatrix}$

$R_3 = R_3 + R_2 \begin{bmatrix} 2 & -5 & 8 \\ 0 & -12 & 9 \\ 0 & 0 & 0 \end{bmatrix}$

$Aug\ (A|b) = \begin{bmatrix} 2 & -5 & 8 & 0 \\ 0 & -12 & 9 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$Rank(A|b) = 2 \quad Rank = 2$

If its same its consistant

Non trivial

2) $x_1 - 3x_2 + 7x_3 = 0$

$-2x_1 + x_2 - 4x_3 = 0$

$x_1 + 2x_2 + 9x_3 = 0$

$$\begin{bmatrix} 1 & -3 & 7 \\ -2 & 1 & -4 \\ 1 & 2 & 9 \end{bmatrix}$$

$R_2 = R_2 + 2R_1$

$$\begin{bmatrix} 1 & -3 & 7 \\ 0 & -5 & 10 \\ 1 & 2 & 9 \end{bmatrix}$$

$R_3 = R_2 - R_3$

$$\begin{bmatrix} 1 & -3 & 7 \\ 0 & -5 & 10 \\ 0 & 5 & 2 \end{bmatrix}$$

$R_3 = R_3 + R_2$

$$\begin{bmatrix} 1 & -3 & 7 \\ 0 & -5 & 10 \\ 0 & 0 & 12 \end{bmatrix}$$

$Rank(A) = 3$

$$A|b = \begin{bmatrix} 1 & -3 & 7 & 0 \\ 0 & -5 & 10 & 0 \\ 0 & 0 & 12 & 0 \end{bmatrix}$$

$Rank(A|b) = 3$

Trivial.

3) $-3x_1 + 5x_2 - 7x_3 = 0$
$-6x_1 + 7x_2 + x_3 = 0$

$$A = \begin{bmatrix} -3 & 5 & -7 \\ -6 & 7 & 1 \end{bmatrix}$$

$R_2 = R_2 - 2R_1$ $\begin{bmatrix} -3 & 5 & -7 \\ 0 & -3 & 15 \end{bmatrix}$

$Rank(A) = 2$
$Rank(A|b) = 2$

$$A|b = \begin{bmatrix} -3 & 5 & -7 & 0 \\ 0 & -3 & 15 & 0 \end{bmatrix}$$

Colum ≠ Rank
non-trivial:

4) $-5x_1 + 7x_2 + 9x_3 = 0$
$x_1 - 2x_2 + 6x_3 = 0$

$$A = \begin{bmatrix} -5 & 7 & 9 \\ 1 & -2 & 6 \end{bmatrix}$$

$R_2 = R_2 + \frac{1}{5} R_1$ $\begin{bmatrix} -5 & 7 & 9 \\ 0 & \frac{33}{5} & -\frac{21}{5} \end{bmatrix}$

Q) Solve the following system of eqn?

$$3x_1 + 2x_2 - 5x_3 = 2$$
$$x_1 + x_2 - 2x_3 = 1$$
$$x_1 - 3x_3 = 4$$

$$A = \begin{bmatrix} 3 & 2 & -5 \\ 1 & 1 & -2 \\ 1 & 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$$

~~$R_2 = R_2 - \frac{1}{3}R_1$~~ ~~$\begin{bmatrix} 3 & 2 & -5 \\ 0 & & \end{bmatrix}$~~

$$A = \begin{bmatrix} 1 & 1 & -2 \\ 3 & 2 & -5 \\ 1 & 0 & -3 \end{bmatrix}$$

$R_2 = R_2 - 3R_1$ $\begin{bmatrix} 1 & 1 & -2 \\ 0 & -1 & 1 \\ 1 & 0 & -3 \end{bmatrix}$

$R_3 = R_3 - R_1$ $\begin{bmatrix} 1 & 1 & -2 \\ 0 & -1 & 1 \\ 0 & -1 & -1 \end{bmatrix}$

$R_3 = R_3 - R_2$ $\begin{bmatrix} 1 & 1 & -2 \\ 0 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix}$

Aug = A|b

$$A = \begin{bmatrix} 3 & 2 & -5 & 2 \\ 1 & 1 & -2 & 1 \\ 1 & 0 & -3 & 4 \end{bmatrix}$$

$-2+\frac{5}{3}$

-6

$1-\frac{2}{3}$

$4-\frac{2}{3}$

$\frac{12-2}{3}$

$R_2 = R_2 - \frac{1}{3}R_1$

$$\begin{bmatrix} 3 & 2 & -5 & 2 \\ 0 & 1/3 & -1/3 & 1/3 \\ 1 & 0 & -3 & 4 \end{bmatrix}$$

$R_3 = R_3 - \frac{1}{3}R_1$

$$\begin{bmatrix} 3 & 2 & -5 & 2 \\ 0 & 1/3 & -1/3 & 1/3 \\ 0 & -2/3 & -4/3 & 10/3 \end{bmatrix}$$

$R_3 = R_3 + 2R_2$

$$\begin{bmatrix} 3 & 2 & -5 & 2 \\ 0 & 1/3 & -1/3 & 1/3 \\ 0 & 0 & -2 & 4 \end{bmatrix}$$

$Ax = b$

$$\begin{bmatrix} 3 & 2 & -5 \\ 0 & 1/3 & -1/3 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1/3 \\ 4 \end{bmatrix}$$

$3x_1 + 2x_2 - 5x_3 = 2$

$1/3\,x_2 - 1/3\,x_3 = 1/3$

$-2x_3 = 4$

$x_3 = -2$ //

$\frac{1}{3}x_2 + \frac{2}{3} = \frac{1}{3}$

$\frac{1}{3}x_2 = -\frac{1}{3}$

$x_2 = -1$ //

$3x_1 - 4 + 10 = 2$

$3x_1 = -4$

$x_1 = -\frac{4}{3}$ //

$3x_1 - 2 + 10 = 2$

$3x_1 = -6$

$x_1 = -2$

b) $x + y + z = 8$

$2x + 3y - z = -2$

$3x + 2y - 9z = 9$

$$A/b = \begin{bmatrix} 1 & 1 & 1 & 8 \\ 2 & 3 & -1 & -2 \\ 3 & 2 & -9 & 9 \end{bmatrix}$$

$R_2 = R_2 - 2R_1$

$$\begin{bmatrix} 1 & 1 & 1 & 8 \\ 0 & 1 & -3 & -18 \\ 3 & 2 & -9 & 9 \end{bmatrix}$$

$R_3 = R_3 - 3R_1$

$$\begin{bmatrix} 1 & 1 & 1 & 8 \\ 0 & 1 & -3 & -18 \\ 0 & -1 & -12 & -15 \end{bmatrix}$$

$R_3 = R_3 + R_2$

$$\begin{bmatrix} 1 & 1 & 1 & 8 \\ 0 & 1 & -3 & -18 \\ 0 & 0 & -15 & -33 \end{bmatrix}$$

$24 - 9$

$x + y + z = 8$

$y - 3z = -18$

$-15z = -33$

$z = \frac{-33}{-15} = \frac{11}{5}$

$y - \frac{33}{5} = -18$

$y = -18 + \frac{33}{5}$

$y = \frac{-90 + 33}{5} = \frac{-57}{5}$

$x - \frac{57}{5} + \frac{11}{5} = 8$

$x - \frac{46}{5} = 8$

$x = 8 + \frac{46}{5}$

$= \frac{86}{5}$

Solve:

$$\begin{bmatrix} 0 & 1 & 5 \\ 1 & 4 & 8 \\ 2 & 7 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ -2 \\ -1 \end{bmatrix}$$

$$A|b = \begin{bmatrix} 0 & 1 & 5 & -4 \\ 1 & 4 & 8 & -2 \\ 2 & 7 & 1 & -1 \end{bmatrix}$$

$$R_2 = R_2 \begin{bmatrix} 1 & 4 & 8 & -2 \\ 0 & 1 & 5 & -4 \\ 2 & 7 & 1 & -1 \end{bmatrix}$$

$$R_3 = R_3 - 2R_1 \begin{bmatrix} 1 & 4 & 8 & -2 \\ 0 & 1 & 5 & -4 \\ 0 & -1 & -5 & -1 \end{bmatrix}$$

$$R_3 = R_3 + R_2 \begin{bmatrix} 1 & 4 & 8 & -2 \\ 0 & 1 & 5 & -4 \\ 0 & 0 & 0 & -5 \end{bmatrix}$$

→ A Pivot pos in a matrix A is a loc in A that correspond to the leading entry in the row echlon form of A.

→ A Pivot column is a column in A that contains the pivot position.

→ A variable corresponds to a non-pivot column is called a free variable.

Solve

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$$A|b = \begin{bmatrix} 2 & 1 & 1 & 2 \\ 1 & -1 & 2 & -2 \end{bmatrix}$$

~~$R_2 = R_1 - 2R_2$~~ $\begin{bmatrix} 2 & 1 & 1 & 2 \\ 0 & 3 & -3 & 6 \end{bmatrix}$

$R_2 - \frac{1}{2}R_1 \begin{bmatrix} 2 & 1 & 1 & 2 \\ 0 & -\frac{3}{2} & \frac{13}{2} & -3 \end{bmatrix}$

$\text{Rank}(A|b) = \text{Rank}(A) = 2$

~~2 sol~~

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & -\frac{3}{2} & \frac{3}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

$2x_1 + x_2 + x_3 = 2$

$-\frac{3}{2}x_2 + \frac{3}{2}x_3 = -3$

$x_3 = t$

$-\frac{3}{2}x_2 + \frac{3}{2}t = -3.$

$-\frac{3}{2}x_2 = -3 - \frac{3}{2}t$

$-3x_2 = -6 - 3t$

$x_2 = 2 + t$

~~$-\frac{3}{2}x_2 = -6 - 3t$~~

~~$-3x_2 = -9t$~~

~~$x_2 = 3t$~~

~~$2x_1 + 4t = 2$~~

~~$2x_1 = 2 - 4t$~~

~~$x_1 = 1 - 2t$~~

$2x_1 + 2 + t + t = 2$

$2x_1 + 2 + 2t = 2$

$x_1 = -t$

Ex:

$$\text{Red}(A|b) = \left[\begin{array}{ccccc|c} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{array}\right]$$

~~$x_1 + 6x_2$~~ $x_2 \; x_4$ - free variable
$\downarrow t \quad \downarrow s$

$$x_1 + 6t + 2x_3 - 5s - 2x_5 = -4$$
$$2x_3 - 8s - x_5 = 3$$
$$x_5 = 1$$

$$2x_3 - 8s - 1 = 3$$
$$2x_3 + 8s = 4$$
$$x_3 - 4s = 2$$
$$x_3 = 2 + 4s$$

$$x_1 + 6t + 4 + 8s - 5s - 2 = -4$$
$$x_1 + 6t + 3s = -6$$
$$x_1 = -6t - 3s - 6$$

H.W

7. $\begin{bmatrix} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{bmatrix}$

$x_1 + 3x_2 + 4x_3 = 7$

$3x_1 + 9x_2 + 7x_3 = 6$

$R_2 = R_2 - 3R_1 \begin{bmatrix} 1 & 3 & 4 & 7 \\ 0 & 0 & -5 & -15 \end{bmatrix}$

free Variable

$x_1 + 3x_2 + 4x_3 = 7$

$-5x_3 = -15$

$x_2 = 3$

$x_3 = t$

$x_1 + 3t + 12 = 7$

$x_1 = -5 - 3t$

8) $\begin{bmatrix} 1 & 4 & 0 & 7 \\ 2 & 7 & 0 & 10 \end{bmatrix}$

$R_2 = R_2 - 2R_1 \begin{bmatrix} 1 & 4 & 0 & 7 \\ 0 & -1 & 0 & -4 \end{bmatrix}$

t

$x_1 + 4x_2 = 7$

$-x_2 = -4$

$x_2 = 4$

$x_3 = t$

$x_1 = 7 - 16 =$

$x_1 = -9$

9) $\begin{bmatrix} 0 & 1 & -6 & 5 \\ 1 & -2 & 7 & -6 \end{bmatrix}$

$R_1 \leftrightarrow R_2 \begin{bmatrix} 1 & -2 & 7 & -6 \\ 0 & 1 & -6 & 5 \end{bmatrix}$

t

$x_1 - 2x_2 + 7x_3 = -6$

$x_2 - 6x_3 = 5$

$x_3 = t$

$x_2 = 5 + 6t$

$x_1 - 10 - 12t + 7t = -6$

$x_1 - 10 - 5t = -6$

$x_1 = 4 + 5t$

10) $\begin{bmatrix} 1 & -2 & -1 & 3 \\ 3 & -6 & -2 & 2 \end{bmatrix}$

$R_2 = R_2 - 3R_1 \begin{bmatrix} 1 & -2 & -1 & 3 \\ 0 & 0 & -1 & -7 \end{bmatrix}$

t

$x_1 - 2t - x_3 = 3$

$x_3 = -7$

$x_1 - 2t + 7 = 3$

$x_1 = 2t - 4$

$x_2 = t$

11) $\begin{bmatrix} 3 & -4 & 2 & 0 \\ -9 & 12 & -6 & 0 \\ -6 & 8 & -4 & 0 \end{bmatrix}$

$R_2 = R_2 + 3R_1$ $\begin{bmatrix} 3 & -4 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ -6 & 8 & -4 & 0 \end{bmatrix}$

$R_3 = R_3 + 2R_1$ $\begin{bmatrix} 3 & -4 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$s \quad t$

$3x_1 - 4s + 2t = 0$

$x_1 = \frac{-2t}{3} + \frac{4s}{3}$

$x_2 = s \quad x_3 = t$

12) $\begin{bmatrix} 1 & -7 & 0 & 6 & 5 \\ 0 & 0 & 1 & -2 & -3 \\ -1 & 7 & -4 & 2 & 7 \end{bmatrix}$

$R_3 = R_3 + R_1$ $\begin{bmatrix} 1 & -7 & 0 & 6 & 5 \\ 0 & 0 & 1 & -2 & -3 \\ 0 & 0 & -4 & 8 & 12 \end{bmatrix}$

$R_3 = R_3 + 4R_2$ $\begin{bmatrix} 1 & -7 & 0 & 6 & 5 \\ 0 & 0 & 1 & -2 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$s \quad t$

$x_3 - 2t = -3$

$x_3 = 2t - 3$

$x_1 - 7s + 6t = 5; \quad x_1 = 7s - 6t + 5$

$x_2 = s$

$x_4 = t$

13)
$$\left[\begin{array}{cccccc} 1 & -3 & 0 & -1 & 0 & -2 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$
s t

$x_4 + 9t = 4$

$x_4 = 4 - 9t$

$x_3 = s \qquad x_5 = t$

$x_2 - 4t = 1$

$x_2 = 1 + 4t$

$x_1 - 3 - 12t - 4 + 9t = -2$

$x_1 - 7 - 3t = -2$

$x_1 = 5 + 3t$

14)
$$\left[\begin{array}{cccccc} 1 & 2 & -5 & -6 & 0 & -5 \\ 0 & 1 & -6 & -3 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$
s t

$x_3 = t \qquad x_3 = s \qquad x_5 = 0$

$x_2 - 6s - 3t = 2$

$x_2 = 2 + 6s + 3t$

$x_1 + 4 + 12s + 6t - 5s - 6t = -5$

$x_1 + 7s = -9$

$x_1 = -7s - 9$

Q. Find an eq^n involving g, h & K that makes this augmented matrice corresponds to a consistant system.

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ -2 & 5 & -9 & K \end{bmatrix}$$

$$x_1 - 4x_2 + 7x_3 = g$$
$$3x_2 - 5x_3 = h$$
$$-2x_1 + 5x_2 - 9x_3 = K$$

$R_3 = R_3 + 2R_1$

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ 0 & -3 & 5 & K+2g \end{bmatrix}$$

$R_3 = R_3 + R_2$

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ 0 & 0 & 0 & K+2g+h \end{bmatrix}$$

$K+2g+h = 0$ then consistant.

Q. Find the value of h, K

$$x + 3y - 2z = -1$$
$$2x + 5y + z = 2$$
$$2x + 6y - hz = K$$

$$\begin{bmatrix} 1 & 3 & -2 & -1 \\ 2 & 5 & 1 & 2 \\ 2 & 6 & -h & K \end{bmatrix}$$

$R_2 = R_2 - 2R_1$

$$\begin{bmatrix} 1 & 3 & -2 & -1 \\ 0 & -1 & 5 & 4 \\ 2 & 6 & -h & K \end{bmatrix}$$

$R_3 = R_3 - 2R_1$

$$\begin{bmatrix} 1 & 3 & -2 & -1 \\ 0 & -1 & 5 & 4 \\ 0 & 0 & -h+4 & K+2 \end{bmatrix}$$

no sol^n a) $h=4$ $k\neq -2$ $[\text{Rank}(A|b) \neq \text{Rank}(A)]$

inf b) $h=4$ $k=-2$ $[\text{Rank}(A|b) = \text{Rank}(A)] \neq 3$

unique c) $h\neq 4$ $k\in\mathbb{R}$ $[\text{Rank}(A|b) = \text{Ra}_n(A)] = 3$

H.W

1.2: 17, 18, 19, 20

17) $\begin{bmatrix} 2 & 3 & h \\ 4 & 6 & 7 \end{bmatrix}$

$R_2 \to R_2 - 2R_1$ $\begin{bmatrix} 2 & 3 & h \\ 0 & 0 & 7-2h \end{bmatrix}$

$7-2h=0$

$7=2h$

$h=\frac{7}{2}$

18) $\begin{bmatrix} 1 & -3 & -2 \\ 5 & h & -7 \end{bmatrix}$

$R_2 = R_2 - 5R_1$ $\begin{bmatrix} 1 & -3 & -2 \\ 0 & h+15 & 3 \end{bmatrix}$

$h \neq -15$

19) $\begin{bmatrix} 1 & h & 2 \\ 4 & 8 & K \end{bmatrix}$

$R_2 = R_2 - 4R_1$ $\begin{bmatrix} 1 & h & 2 \\ 0 & 8-4h & K-8 \end{bmatrix}$

a) no sol^n

$8 = 4h$

$h = 2$

$K - 8 \neq 0$

$K \neq 8$

c) inf

$8 - 4h = 0$

$h = 2$

$K - 8 = 0$

$K = 8$

b) unique sol^n

$8 - 4h \neq 0$

$h \neq 2$

$K \in R$

20 $\begin{bmatrix} 1 & 3 & 2 \\ 3 & h & K \end{bmatrix}$

$R_2 = R_2 - 3R_1$ $\begin{bmatrix} 1 & 3 & K \\ 0 & h-9 & K-3K \end{bmatrix}$ $\begin{bmatrix} 1 & 3 & 2 \\ 0 & h-9 & K-6 \end{bmatrix}$

a) no sol^n

$h - 9K = 0$

$h = 9K$

$K - 3K$

$h - 9 = 0$

$h = 9$

$K - 6 \neq 0$

$K \neq 6$

b) unique sol^n

$h - 9 \neq 0$

$h \neq 9$

$K \in R$

c) inf

$h - 9 = 0$

$h = 9$

$K - 6 = 0$

$K = 6$

Vector Eqn

$$Ax = b$$

$$\begin{bmatrix} a_{11} & a_{12} \cdots a_{1n} \\ a_{21} & a_{22} \cdots a_{2n} \\ \vdots & \\ a_{m1} & a_{m2} \cdots a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$\begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} x_1 + \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} x_2 + \cdots + \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} x_n = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

linear combination of vector $\{v_1, v_2, \ldots v_n\}$

$$= \{c_1v_1 + c_2v_2 + \cdots + c_nv_n : c_1, c_2, \ldots c_n \in \mathbb{R}\}$$

VECTOR SPACE

A vector space is a non empty set V of objects, called vectors, on which are defined two operations called addition and multiplication by scalars (real numbers) subject to the ten axioms (or rules) listed below. The axioms must hold for all vectors u, v, and w in V and for all scalars c & d.

1. The sum of u & v, denoted by u+v, is in V.
2. $u+v = v+u$
3. $(u+v)+w = u+(v+w)$

4. There is a zero vector 0 in V such that $u+0=u$
5. For each u in V, there is a vector $-u$ in V such that $u+(-u)=0$
6. The scalar multiple of u by c, denoted by cu, is in V.
7. $c(u+v)=cu+cv$
8. $(c+d)u=cu+du$
9. $c(du)=(cd)u$
10. $1u=u$

Example

① Set of all real number = V (under usual ddd & mul)

②

Subspace = Subset + Vector space

Only subspaces of $\mathbb{R}$ - $\{0\} \subset \mathbb{R}$
$\mathbb{R} \subseteq \mathbb{R}$
and is called trivial subspace.

4.1

1a) $u = x_1, y_1$ $\quad v = x_2, y_2$
$x_1+x_2 \geq 0$ $\quad y_1+y_2 \geq 0$ as $x_1, x_2, y_1, y_2 \geq 0$
so its closed.

b) ~~c≠~~ $c=-1$

2a) $(cx)(cy) = \underline{c^2xy \geq 0}$

b) $u=\begin{bmatrix}2\\1\end{bmatrix}$ $\quad v=\begin{bmatrix}-1\\-2\end{bmatrix}$ $\quad u+v=\begin{bmatrix}1\\-1\end{bmatrix} \notin W$

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3) $u = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $c = 2$ $2u = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \notin H$

5) Yes [Closed under all axioms]
6) No [not closed for scalar multiply]
7) No [not for scalar multiply]
8) Yes [Closed under all axioms]

Subspace Test

① Zero element
② Closed under add
③ Closed under mul.

Vector Space

exc: Set of all polynomial deg $\leq n$ → Yes
Set of all polynomial deg with 5 → No.
Set of all mxn matrix with real entries → Yes
~~$\mathbb{R}^2 \in$~~
$\mathbb{R}^2 \subseteq \mathbb{R}^3$
$\mathbb{R}^2 = \{(x,y) : x, y \in \mathbb{R}\}$
$\mathbb{R}^3 = \{(x,y,z) : x, y, z \in \mathbb{R}\}$

$S = \{(x,y,0) : x, y \in \mathbb{R}\}$ xy plane
$S = \{(x,0,z) : x, z \in \mathbb{R}\}$ xz plane
$S = \{(0,y,z) : y, z \in \mathbb{R}\}$ yz plane.

Q Is $v = \begin{pmatrix} 19 \\ 10 \\ -1 \end{pmatrix}$ is in the span of (S) where S:

$$S = \left\{ \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} -5 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 7 \\ -4 \end{pmatrix} \right\}$$

Span = Set of all linear combination of (v_1, v_2, v_3)

$$\begin{pmatrix} 19 \\ 10 \\ -1 \end{pmatrix} = C_1 \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} + C_2 \begin{pmatrix} -5 \\ 0 \\ 1 \end{pmatrix} + C_3 \begin{pmatrix} 1 \\ 7 \\ -4 \end{pmatrix}$$

$19 = 3C_1 - 5C_2 + C_3$

$10 = -C_1 + 7C_3$

$-1 = 2C_1 + C_2 - 4C_3$

$-1 = 2(7C_3 - 10) + C_2 - 4C_3$

$-1 = 14C_3 - 20 + C_2 - 4C_3$

$-1 = 10C_3 - 20 + C_2$

$19 = 21C_3 - 30 - 5C_2 + C_3$

$19 = 22C_3 - 30 - 5C_2$

$49 = 22C_3 - 5C_2$

$19 = 10C_3 + C_2$

$$\begin{bmatrix} 3 & -5 & 1 \\ -1 & 0 & 7 \\ 2 & 1 & -4 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \\ C_3 \end{bmatrix} = \begin{bmatrix} 19 \\ 10 \\ -1 \end{bmatrix}$$

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Q. $v = \begin{pmatrix} 3 \\ -2 \\ -4 \\ 1 \end{pmatrix}$ $S = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ -3 \\ 1 \end{pmatrix} \right\}$

$$\begin{bmatrix} 1 & 1 & 2 & 3 \\ 2 & -1 & 0 & -2 \\ 3 & 1 & -3 & -4 \\ 4 & -1 & 1 & 1 \end{bmatrix}$$

$R_2 = R_2 - 2R_1$

$$\begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & -3 & -4 & -8 \\ 3 & 1 & -3 & -4 \\ 4 & -1 & 1 & 1 \end{bmatrix}$$

$R_3 = R_3 - 3R_1$

$$\begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & -3 & -4 & -8 \\ 0 & -8 & 9 & 20 \\ 4 & -1 & 1 & 1 \end{bmatrix}$$

$R_3 = R_3 + \frac{8}{3}R_2$

$$\begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & -3 & -4 & -8 \\ 0 & 0 & 20 & \frac{136}{3} \\ 4 & -1 & 1 & 1 \end{bmatrix}$$

$R_4 = R_4 - 4R_1$

$$\begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & -3 & -4 & -8 \\ 0 & 0 & 20 & \frac{136}{3} \\ 0 & -5 & -7 & -11 \end{bmatrix}$$

$\frac{136}{24}$

$-3x = -8$

$x = \frac{8}{3}$

$-8 \mid \frac{160}{3}$

$\frac{-24+160}{3}$

Find the span of $\left\{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}\right\}$ in $\mathbb{R}^2$.

$$S = \left\{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}\right\}$$

$$= \left\{\begin{pmatrix}c_1\\c_2\end{pmatrix} : c_1, c_2 \in \mathbb{R}\right\}$$

Q. Determine weather $\left\{\begin{pmatrix}1\\2\end{pmatrix}\begin{pmatrix}3\\4\end{pmatrix}\right\}$ span $\mathbb{R}^2$

$$c_1\begin{pmatrix}1\\2\end{pmatrix} + c_2\begin{pmatrix}3\\4\end{pmatrix}$$

$$c_1 + 3c_2$$

$$2c_1 + 4c_2$$

$$\left\{\begin{pmatrix}c_1+3c_2\\2c_1+4c_2\end{pmatrix} : c_1, c_2 \in \mathbb{R}\right\}$$

N.W Determine weather $\left\{\begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}2\\4\end{pmatrix}\right\}$ span $\mathbb{R}^2$

$$\begin{bmatrix}1 & 2\\2 & 4\end{bmatrix}$$

Row

$$R_2 = R_2 - 2R_1 \begin{bmatrix}1 & 2\\0 & 0\end{bmatrix}$$

Row(A) ≠ no of columns

hence no.

$$\{(1,2), (2,4)\} = \{(x,y) : y = 2x\} \neq \mathbb{R}^2$$

H.W

1) $\begin{bmatrix} 8 & -4 & -7 & 9 \\ -4 & 3 & 6 & -4 \\ -3 & -2 & -5 & -4 \\ 9 & -8 & -18 & 7 \end{bmatrix}$

~~$R_2 = R_2 + 2R$~~ $\begin{bmatrix} -4 & 3 & 6 & -4 \\ 8 & -4 & -7 & 9 \\ -3 & -2 & -5 & -4 \\ 9 & -8 & -18 & 7 \end{bmatrix}$

$R_2 = R_2 + 2R_1$ $\begin{bmatrix} -4 & 3 & 6 & -4 \\ 0 & 2 & 5 & -1 \\ -3 & -2 & -5 & -4 \\ 9 & -8 & -18 & 7 \end{bmatrix}$

$\begin{bmatrix} 1 & -1/2 & 0 & 7/8 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & -4 \end{bmatrix}$

not a substabe

22) Let $F = \begin{bmatrix} a_1 & d_1 \\ b_1 & e_1 \\ c_1 & f_1 \end{bmatrix}_{3\times 2}$.

$$M = \left\{ \begin{bmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \end{bmatrix} \mid FA = 0 \right\}$$

Q.a) Verify weather the set $\{(1,3,3), (0,1,2), (-2,0,1)\}$ are LI.

b) Verify weather span of $S = \mathbb{R}^3$ ::

$$C_1 - 2C_3 = x$$
$$2C_1 + C_2 = y$$
$$3C_1 + 2C_2 + C_3 = z$$

$$\begin{bmatrix} 1 & 0 & -2 & x \\ 2 & 1 & 0 & y \\ 3 & 2 & 1 & z \end{bmatrix}$$

$R_2 = R_2 - 2R_1$

$$\begin{bmatrix} 1 & 0 & -2 & x \\ 0 & 1 & 4 & y-2x \\ 3 & 2 & 1 & z \end{bmatrix}$$

$R_3 = R_3 - 3R_1$

$$\begin{bmatrix} 1 & 0 & -2 & x \\ 0 & 1 & 4 & y-2x \\ 0 & 2 & 5 & z-3x \end{bmatrix}$$

$R_3 = R_3 - 2R_2$

$$\begin{bmatrix} 1 & 0 & -2 & x \\ 0 & 1 & 4 & y-2x \\ 0 & 0 & -3 & z-2y+x \end{bmatrix}$$

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Let V be a vector space and if
$B = \{v_1, v_2, \ldots, v_n\}$ be a subset of V
B is said to be a basis if it satisfies the following condition.

① $\text{Span}(B) = V$
② B is L.I.

The number of elements in B is called dimension of V.

Let $v \in V$

$V = \text{Span}(B)$

$V = c_1v_1 + c_2v_2 + \cdots + c_nv_n$ -①

$V = b_1v_1 + b_2v_2 + \cdots + b_nv_n$ -②

1 - 2

$0 = (c_1 - b_1)v_1 + (c_2 - b_2)v_2 + \cdots + (c_n - b_n)v_n$

$c_1 = b_1$

$c_2 = b_2$

$c_n = b_n$

$c_i = b_i \quad (i \in 1, 2, 3, \ldots, n)$

$(c_1, c_2, \ldots, c_n)$ = Coordinate vector.

Q. Let $V=(1,1,1)$ and

$B=\{(1,2,3),(0,1,2),(-2,0,1)\}$

be a basis of R^3 find the coordinate vector for of V wrt B.

$$(1,1,1) = C_1(1,2,3) + C_2(0,1,2) + C_3(-2,0,1)$$

$1 = C_1 - 2C_3 \quad \Rightarrow C_1 = 1+2C_3$

$1 = 2C_1 + C_2$

$1 = 3C_1 + 2C_2 + C_3$

$C_1 = 1$

$C_2 = -1$

$C_3 = 0$

$$\begin{bmatrix} 1 & 0 & -2 & 1 \\ 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 1 \end{bmatrix}$$

$R_2 = R_2 - 2R_1$

$$\begin{bmatrix} 1 & 0 & -2 & 1 \\ 0 & 1 & 4 & -1 \\ 3 & 2 & 1 & 1 \end{bmatrix}$$

$R_3 = R_3 - 3R_1$

$$\begin{bmatrix} 1 & 0 & -2 & 1 \\ 0 & 1 & 4 & -1 \\ 0 & 2 & 5 & -2 \end{bmatrix}$$

$R_3 = R_3 - 2R_2$

$$\begin{bmatrix} 1 & 0 & -2 & 1 \\ 0 & 1 & 4 & -1 \\ 0 & 0 & -3 & 0 \end{bmatrix}$$

Tutorial

4.1

23) a) If a function in the vector space V of all real valued functions on $\mathbb{R}$ and if $f(t) = 0$ for some t the f is the zero vector in V.
→ False, A function can be 0 at one point without being the zero f^n.

b) A vector is an arrow in 3D space.
→ False, It can be a matrix, function anything

c) A Subset H of a vector space V is a subspace of V if the zero vector is in H.
→ False, must be also closed under addition and multiplication.

d) A subspace is also a vector space.
→ True, subspace satisfies all the axioms of its vector space.

24) a) A vector is any element of vector space.
→ True

b) If u is a vector in a vector space V, the $-1(u)$ is the same as negative of u.
→ True, by the property of scalar multiplication.

c) A vector space is also a subspace.

→ True, A vector space is a subspace of itself as it satisfies all conditions.

d) R^2 is a subspace of R^3.

→ False, R^2 is not even subset of R^3 as R^2 has 2 elements and R^3 has 3.

2.2

29) $A = \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix}$

$$[AI] = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 4 & 7 & 0 & 1 \end{bmatrix}$$

$R_2 = R_2 - 4R_1 \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & -1 & -4 & 1 \end{bmatrix}$

$R_2 = -R_2 \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 4 & -1 \end{bmatrix}$

$R_1 = R_1 - 2R_2 \begin{bmatrix} 1 & 0 & -7 & 2 \\ 0 & 1 & 4 & -1 \end{bmatrix}$

$$A^{-1} = \begin{bmatrix} -7 & 2 \\ 4 & -1 \end{bmatrix}$$

32) $\begin{bmatrix} 1 & -2 & 1 \\ 4 & -7 & 3 \\ -2 & 6 & -4 \end{bmatrix}$

$[AI] = \begin{bmatrix} 1 & -2 & 1 & 1 & 0 & 0 \\ 4 & -7 & 3 & 0 & 1 & 0 \\ -2 & 6 & -4 & 0 & 0 & 1 \end{bmatrix}$

$R_2 = R_2 - 4R_1 \begin{bmatrix} 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -4 & 1 & 0 \\ 0 & 2 & -2 & 2 & 0 & 1 \end{bmatrix}$

$R_3 = R_3 - 2R_2 \begin{bmatrix} 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -4 & 1 & 0 \\ 0 & 0 & 0 & 10 & -2 & 1 \end{bmatrix}$

This can't be converted to identity as the 3rd row just three element became 0. hence can't compute inverse.

1.6

3)

	C	F	M
C	0.2	0.8	0.4
F	0.3	0.1	0.4
M	0.5	0.1	0.2

$A \begin{bmatrix} 0.2 & 0.8 & 0.4 \\ 0.3 & 0.1 & 0.4 \\ 0.5 & 0.1 & 0.2 \end{bmatrix}$

b) $0.2x + 0.8y + 0.4z = x$

$0.3x + 0.1y + 0.4z = y$

$0.5x + 0.1y + 0.2z = z$

$-0.8x + 0.8y + 0.4z = 0$

$0.3x - 0.9y + 0.4z = 0$

$0.5x + 0.1y - 0.8y = 0$

~~M.W~~

Verify if $\{(1,0,0), (0,1,0), (0,0,1)\}$ is a basis of $\mathbb{R}^3$

$$c_1(1,0,0) + c_2(0,1,0) + c_3(0,0,1) = 0,0,0$$

$$c_1, c_2, c_3 = 0,0,0$$

$$c_i = 0 \text{ for } (i = 1,2,3)$$

Basis = Maximal LI in V

= Minimum Span

~~Find the~~

Q. Find a basis for P_n Determine its dimension.

Q. Find the basis for $M_{2\times 3}(\mathbb{R})$ ".

1. P_n : Set of all polynomial with degree $\leq n$.

let $a_0 + a_1x^1 + a_2x^2 + \dots + a_nx^n$

$B = \{1, x, x^2, \dots, x^n\} \subset P_n$

$C_1 + C_2x + C_3x^2 + \dots + C_{n+1}x^n = 0$

$= 0 + 0x + 0x^2 + \dots + 0x^n = 0$

$C_i = 0$ hence linearly independent.

as B spans P_n as it is the linear combination of P_n.

Hence 'B' is a basis of P_n.

No of element in B is $(n+1)$ hence the dimension is $(n+1)$.

B is also the standard basis of B.

2. $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$

$a_{11}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + a_{12}\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + a_{13}\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} + \dots + a_{23}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\text{Basis} = \left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$

Dimension of $M_{m\times n}(\mathbb{R}) = mn$

Q. Find a basis for the following subspaces of R^3.

a) $\{(a,b,c,d) : a=b=c=d\}$

Ans: $\{(a,a,a,a) : a \in \mathbb{R}\}$

$(a,a,a,a) = a(1,1,1,1)$

Basis $= \{(1,1,1,1)\}$

b) $\{(a,b,a-b,a+b) : a,b \in \mathbb{R}$

$(a,b,a-b,a+b) = a(1,0,1,1) + b(0,1,-1,1)$

Basis $= \{(1,0,1,1), (0,1,-1,1)\}$

Q. Prove that the following are subspaces of R^4.

a) $W = \{(a,b,c,d) : a=b=c=d\}$

i) Zero vector

$a=0$

So, $(0,0,0,0)$ where $a \in R$

$0=0=0=0$

Hence W contains zero vector.

ii) Lets take two vectors in W

say, $u = (a,a,a,a)$, $v = (b,b,b,b)$

$u+v = (a+b, a+b, a+b, a+b)$

$u+v \in W$

Hence W is closed under addition.

iii) Let $k \in R$
and $u = (a, a, a, a) \in W$
then $ku = (ka, ka, ka, ka)$
hence $ku \in W$
hence it is closed under multiplication.

b) $W = \{(a, b, a-b, a+b) : a, b \in R$

i) lets take $a = 0, b = 0$
then $(0, 0, (0-0), (0+0)) = (0, 0, 0, 0)$.
Zero vector belongs to W.

ii) $u = \{a, b, a-b, a+b\}$
$v = \{c, d, c-d, c+d\}$
$u+v = \{a+c, b+d, (a-b)+(c-d), (a+b)+(c+d)\}$

$= \{a+c, b+d, (a+c)-(b+d), (a+c)+(b+d)\}$

hence closed under addition.

iii) let $k \in R$
$u = (a, b, a-b, a+b)$
$ku = (ka, kb, ka-kb, ka+kb)$

hence closed under multiplication.

Q. Find the span of $\{(1,8),(2,8)\}$.

Span of $\{(1,8),(2,8)\}$

$\{C_1(1,8) + C_2(2,8) : C_i \in \mathbb{R}\}$

$(x,y) = C_1(1,8) + C_2(2,8)$

$x = C_1 + 2C_2$

$y = 8C_1 + 8C_2$

Q. Find the span $\{(1,8),(2,16)\}$

$C_1(1,8) + C_2(2,16)$

$C_1(1,8) + 2C_2(1,8)$

Q. Verify set $\{(1,8),(2,8),(1,16)\}$ independent or dependent?

$C_1 + 2C_2 + C_3 = 0$

$8C_1 + 8C_2 + 16C_3 = 0$

$$\begin{bmatrix} 1 & 2 & 1 \\ 8 & 8 & 16 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 1 & 0 \\ 8 & 8 & 16 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & -8 & 8 & 0 \end{bmatrix}$$

It is dependent.

$S \subseteq V$

dimension$(V) = n$

$|S|$ Cardinality of S

$|S| < n$ doesn't span V

$|S| = n$ → one L.I / Span. Check either one

$|S| > n$ dependent

Q. $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

Q. Determine weather the following set of vector is linearly independent and span's $\mathbb{R}^3$. If not find the span of the following sets.

a) $\{(2,-4,3),(1,4,-2)\}$

b) $\{(2,-4,3),(16,-32,24)\}$

c) $\{(1,2,3),(4,5,6),(7,8,9)\}$

d) $\{(2,-1,3),(1,4,-2),(3,2,1)\}$

e) $\{(2,-1,3),(1,4,-4),(3,2,1),(16,-32,24)\}$

a) $C_1(2,-4,3) + C_2(1,4,-2) = 0$

$C_1(2,-4,3) = -C_2(1,4,-2)$

$(2,-4,3) = \frac{-C_2}{C_1}(1,4,2)$

we cant find any scalar $\frac{-C_2}{C_1}$ such that $K(1,4,2)$ is $(2,-4,3)$ hence linearly independent.

$\dim(\mathbb{R}^3) = 3$

in order to span $\mathbb{R}^3$, need at least 3 vectors. but here cardinality of set = 2. So it doesn't span $\mathbb{R}^3$.

Subspace spanned by ①

$$\text{Span}(S_1) = \{c_1(2,-4,3) + c_2(1,4,-2) : c_1, c_2 \in \mathbb{R}\}$$

Let $(x,y,z) \in \text{Span}(S_1)$

~~$(x,y,z) =$~~

$$x = 2c_1 + c_2$$
$$y = -4c_1 + 4c_2$$
$$z = 3c_1 - 2c_2$$

$$\begin{bmatrix} 2 & 1 & x \\ -4 & 4 & y \\ 3 & -2 & z \end{bmatrix}$$

$R_3 +$

$-\frac{7}{2}x + 6x = 0$

$6x = \frac{7}{2}$

$x = \frac{7}{12}$

$$\begin{bmatrix} 2 & 1 & x \\ 0 & 6 & y+2x \\ 3 & -2 & z \end{bmatrix}$$

$R_3 = R_3 - \frac{3}{2}R_1$

$-2 - \frac{3}{2}$

$-4 - \frac{3}{2}$

$$\begin{bmatrix} 2 & 1 & x \\ 0 & 6 & y+2x \\ 0 & -7/2 & z - \frac{3}{2}x \end{bmatrix}$$

$R_3 = R_3 + \frac{7}{12}R_2$

~~$+\frac{7}{2} = 1 + x$~~

~~$R_3 = R_3 + \frac{7}{2}R_1$~~

$$\begin{bmatrix} 2 & 1 & x \\ 0 & 6 & y+2x \\ 0 & 0 & \end{bmatrix}$$

~~$z - \frac{3}{2}x + \frac{7}{2}x$~~

~~$\begin{bmatrix} 2 & 1 & x \\ 0 & 6 & y+2x \\ 0 & 0 & z+2x \end{bmatrix}$~~

$z - \frac{3}{2}x + \frac{7}{12}y + \frac{14}{12}x$

$z - \frac{3}{2}x + \frac{7}{12}z - \frac{21}{24}x$ | $\frac{20}{24}x$

$-\frac{1}{3}$ $\frac{7}{6}$ $\frac{-16x}{12} + \frac{14}{12}x$ $\frac{-2x}{12}$

$\frac{12z + 7z}{12} - \frac{62x - 42x}{24}$

Q. $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

R.S.: $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} =$ R.S: $C_1 = 0$

~~$c_1 = 0$~~

C.S: $C_1 + C_2 = 0 \therefore C_1 = -C_2$ C.S = $C_1 = 0$

N.S: N.S =

b) $S = \{(2,-4,3), (16,-32,24)\}$

Linear independence:

$(16,-32,24) = 8(2,-4,3)$

2nd vector is a scalar multiple of 1st hence linearly dependent.

Only two vectors hence cant span $\mathbb{R}^3$.

span = span $(2,-4,3)$ as the other is dependent

$= (2t, -4t, 3t), t \in R$.

Just a line.

c) $S = \{(1,2,3), (4,5,6), (7,8,9)\}$

$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

$R_2 = R_2 - 4R_1 \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{bmatrix}$; $R_3 = R_3 - 7R_1 \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix}$

$R_3 = R_3 - 2R_2$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank (A) = 2

number of vector = 3 & Rank = 2

$2 < 3$ hence dependent.

$$\begin{bmatrix} 1 & 4 & 7 & x \\ 2 & 5 & 8 & y \\ 3 & 6 & 9 & z \end{bmatrix}$$

$R_2 = R_2 - 2R_1$

$$\begin{bmatrix} 1 & 4 & 7 & x \\ 0 & -3 & -6 & y-2x \\ 3 & 6 & 9 & z \end{bmatrix}$$

$R_3 = R_3 - 3R_1$

$$\begin{bmatrix} 1 & 4 & 7 & x \\ 0 & -3 & -6 & y-2x \\ 0 & -6 & -12 & z-3x \end{bmatrix}$$

$R_3 = R_3 - 2R_2$

$$\begin{bmatrix} 1 & 4 & 7 & x \\ 0 & -3 & -6 & y-2x \\ 0 & 0 & 0 & z-3x-2y+4x \end{bmatrix}$$

$$z + x - 2y = 0$$

Span is the plane $x - 2y + z = 0$.

d) $\{(2,-1,3),(1,4,-2),(3,2,1)\}$

$$\begin{bmatrix} 2 & 1 & 3 & x \\ -1 & 4 & 2 & y \\ 3 & -2 & 1 & z \end{bmatrix}$$

$R_2 = R_2 + \frac{1}{2}R_1$

$$\begin{bmatrix} -1 & 4 & 2 & y \\ 2 & 1 & 3 & x \\ 3 & -2 & 1 & z \end{bmatrix}$$

$R_2 = R_2 - 2R_1$

$$\begin{bmatrix} -1 & 4 & 2 & y \\ 0 & -7 & -3 & x-2y \\ 3 & -2 & 1 & z \end{bmatrix}$$

$R_3 = R_3$

$R_2 = R_2 + 2R_1$

$$\begin{bmatrix} -1 & 4 & 2 & y \\ 0 & 9 & 7 & x+2y \\ 3 & -2 & 1 & z \end{bmatrix}$$

$R_3 = R_3 + 3R_1$

$$\begin{bmatrix} -1 & 4 & 2 & y \\ 0 & 9 & 7 & x+2y \\ 0 & 10 & 7 & z+3y \end{bmatrix}$$

$10 = 9x$

$x = \frac{10}{9}$

$R_3 = R_3 - \frac{10}{9}R_2$

$$\begin{bmatrix} -1 & 4 & 2 & y \\ 0 & 9 & 7 & x+2y \\ 0 & 0 & -\frac{11}{9} & z - \frac{10}{9}x - \frac{47}{9}y \end{bmatrix}$$

$7 - \frac{70}{9}$

$\frac{56-70}{9}$

$R_3 = 9R_3$

$$\begin{bmatrix} -1 & 4 & 2 & y \\ 0 & 9 & 7 & x+2y \\ 0 & 0 & -14 & 9z - 10x - 47y \end{bmatrix}$$

$z - 3y - \frac{10}{9}x - \frac{20}{9}y$

$z - \frac{10}{9}x - \frac{27-20}{9}y$

$z - \frac{-10}{9}x - \frac{47}{9}y$

L.I & Span R^3

e) $\{(2,-1,3),(1,4,-4),(3,2,1),(16,-32,24)\}$

Dependent as cardinality (n) > dimension

Lets check for the span of first 3 vectors.

$\{(2,-1,3),(1,4,-4),(3,2,1)\}$

$$\begin{bmatrix} 2 & 1 & 3 \\ -1 & 4 & 2 \\ 3 & -4 & 1 \end{bmatrix} \quad \begin{bmatrix} -1 & 4 & 2 \\ 2 & 1 & 3 \\ 3 & -4 & 1 \end{bmatrix}$$

$R_2 = R_2 + \frac{1}{2}R_1$

$$\begin{bmatrix} -1 & 4 & 2 \\ 0 & 9 & 7 \\ 3 & -4 & 1 \end{bmatrix}$$

$R_3 = R_3 + 3R_1$

$$\begin{bmatrix} -1 & 4 & 2 \\ 0 & 9 & 7 \\ 0 & 8 & 7 \end{bmatrix}$$

$9x = 8$

$x = \frac{8}{9}$

$R_3 = R_3 - \frac{8}{9}R_2$

$$\begin{bmatrix} -1 & 4 & 2 \\ 0 & 9 & 7 \\ 0 & 0 & - \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Fundamental Spaces

1. Row space
2. Column space
3. Null space of $A \rightarrow \{x : Ax = 0\}$
4. Null space of A^T

Rowspace of $A = \{C_1(R_1) + C_2(R_2) + C_3(R_3) : C_1, C_2, C_3 \in R\}$

Col space of $A = \{C_1(Col_1) + C_2(Col_2) + C_3(Col_3) : C_1, C_2, C_3 \in R\}$

Null space of $A = \left\{(x, y, z) : \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}\right\}$

* Technically Row space & Col space is just span of Row vector & Col vector:

eg: Col space

$$\begin{bmatrix} 1 & 4 & 7 & x \\ 2 & 5 & 8 & y \\ 3 & 6 & 9 & z \end{bmatrix}$$

$C_1(1,4,7) + C_2(2,5,8) + C_3(3,6,9)$

$$\begin{bmatrix} 1 & 2 & 3 & x \\ 4 & 5 & 6 & y \\ 7 & 8 & 9 & z \end{bmatrix}$$

$R_2 = R_2 - 4R_1$

$$\begin{bmatrix} 1 & 2 & 3 & x \\ 0 & -3 & -6 & y-4x \\ 7 & 8 & 9 & z \end{bmatrix}$$

$R_3 = R_3 - 7R_1$

$$\begin{bmatrix} 1 & 2 & 3 & x \\ 0 & -3 & -6 & y-4x \\ 0 & -6 & -12 & z-7x \end{bmatrix}$$

$R_3 = R_3 - 2R_2$

$$\begin{bmatrix} 1 & 2 & 3 & x \\ 0 & -3 & -6 & y-4x \\ 0 & 0 & 0 & z-7x-2y+8x \end{bmatrix}$$

$z+x-2y$

Col space: $z - 2y + x = 0$

eg: Null space of A =

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 0 \\ 7 & 8 & 9 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & -3 & -6 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

z = free variable = t

$x = t$

$x + 2y + 3z = 0$

$-3y - 6z = 0$

$-3y = 6t$

$y = -2t$

$x - 4t + 3t = 0$

$x - t = 0$

$x = t$

Q. Find the row echlon form of $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ and find its row space, col space and null space.

Row echlon: $\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix}$

Row space $= \{C_1(1,2,3) + C_2(0,-3,-6) : C_1, C_2 \in \mathbb{R}\}$

$x = C_1$

$y = 2C_1 - 3C_2 \Rightarrow y = 2x - 3C_2$

$z = 3C_1 - 6C_2 \Rightarrow z = 3x - 6C_2$

$\frac{y - 2x}{3} = -C_2$

$\frac{y - 2x}{-3} = -C_2$

$z = 3x - 6\left(\frac{y - 2x}{-3}\right)$

$z = 3x + 2y - 4x$

$x - 2y + z = 0$

Q. Find a basis for the subspace of R^4 span by the vectors ~~{(1,2,3,4),(2,3,4,5),(3,4,5,6),}~~ {(1,2,3,4),(4,5,6,7),(7,8,9,10)}

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 7 & 8 & 9 & 10 \end{bmatrix}$$

$R_2 = R_2 - 4R_1$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -6 & -9 \\ 7 & 8 & 9 & 10 \end{bmatrix}$$

$R_3 = R_3 - 7R_1$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -6 & -9 \\ 0 & -6 & -12 & -18 \end{bmatrix}$$

$R_3 = R_3 - 2R_2$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -6 & -9 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$